2025 AMC 10A Problem 25 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 25. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 25

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Problem:

A point P is chosen at random inside square $ABCD$. The probability that $\overline{AP}$ is neither the shortest nor the longest side of $\triangle APB$ can be written as $\dfrac{a+b\pi-c\sqrt{d}}{e}$, where a,b,c,d,e are positive integers, $\gcd(a,b,c,e)=1$, and d is not divisible by the square of any prime. What is $a+b+c+d+e$?

Answer Choices:

A. 25

B. 26

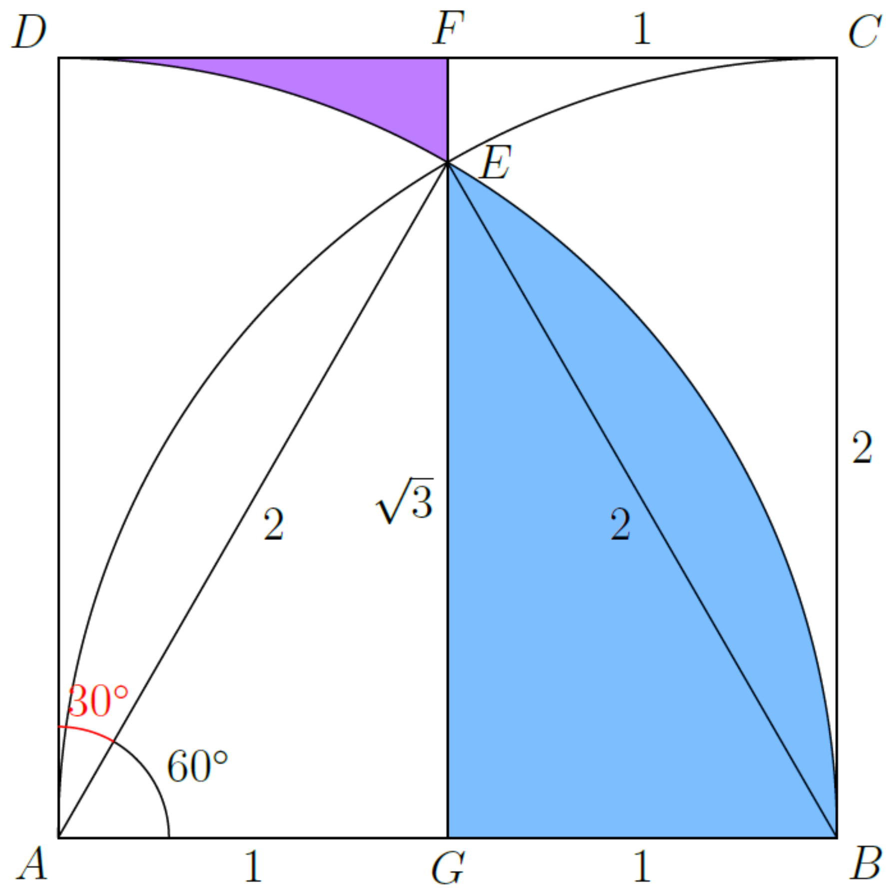
C. 27

D. 28

E. 29

Solution:

Without loss of generality, let the side length of the square $ABCD$ be 2. Draw quarter circles of radius 2 centered at A and B. Let G be the midpoint of AB (so $AG=BG=1$), let F be the midpoint of CD, and let E be the intersection of line FG with the two arcs.



Since $AE = BE = 2$ and $AG = 1$, we have

$$GE = \sqrt{AE^2 - AG^2} = \sqrt{4-1} = \sqrt{3}, \qquad \angle GAE = 60^\circ, \qquad \angle DAE = 30^\circ.$$

We can calculate the area of the blue region as

$$\text{Area of the blue region} = [\triangle GEB] + [\text{sector } AEB] - [\triangle AEB].$$

We compute each part individually

$$[\triangle GEA] = \frac{1}{2} \cdot AG \cdot GE = \frac{1}{2} \cdot 1 \cdot \sqrt{3} = \frac{\sqrt{3}}{2}, \qquad [\triangle GEB] = \frac{\sqrt{3}}{2},$$

$$[\triangle AEB] == \frac{1}{2} \cdot 2\sqrt{3} = \sqrt{3},$$

$$[\text{sector} AEB] = \frac{60^\circ}{360^\circ} \cdot \pi \cdot 2^2 = \frac{1}{6} \cdot 4\pi = \frac{2\pi}{3}.$$

Hence

$$\text{Area of the blue region} = \frac{\sqrt{3}}{2} + \frac{2\pi}{3} - \sqrt{3} = \frac{2\pi}{3} - \frac{\sqrt{3}}{2}.$$

We can compute the area of the purple region as well:

$$\text{Purple} = [AGFD] - [\text{sector } ADE] - [\triangle AGE].$$

Here $AGFD$ is the rectangle with width $AG = 1$ and height $AD = 2$, so $[AGFD] = 2$.
Also,

$$[\text{sector } ADE] = \frac{30^\circ}{360^\circ} \cdot \pi \cdot 2^2 = \frac{1}{12} \cdot 4\pi = \frac{\pi}{3}, \qquad [\triangle AGE] = \frac{\sqrt{3}}{2}.$$

Thus

$$\text{Purple} = 2 - \frac{\pi}{3} - \frac{\sqrt{3}}{2}.$$

$$\textbf{Total desired area} = \text{Pink} + \text{Purple} = \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2}\right) + \left(2 - \frac{\pi}{3} - \frac{\sqrt{3}}{2}\right) = 2 + \frac{\pi}{3} - \sqrt{3}.$$

Since the square has area 4, the probability equals

$$\mathbb{P} = \frac{2 + \frac{\pi}{3} - \sqrt{3}}{4} = \frac{1}{2} + \frac{\pi}{12} - \frac{\sqrt{3}}{4} = \frac{6 + \pi - 3\sqrt{3}}{12}.$$

Therefore, in the form $\dfrac{a + b\pi - c\sqrt{d}}{e}$ we have

$$a = 6, \quad b = 1, \quad c = 3, \quad d = 3, \quad e = 12,$$

so

$$a + b + c + d + e = \boxed{\textbf{(A) } 25}$$